# PrivCap: Benchmarking Cross-Modal Privacy Detection in Vision-Language Models

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## Abstract

Privacy risk in a social-media post depends on both what its image reveals and what its caption states. Existing visual-privacy benchmarks largely evaluate images alone or use text only as an instruction. They therefore cannot isolate the effect of natural caption content or test whether a model attributes a disclosure to the correct modality.

To our knowledge, **PRIVCAP** is the first controlled post-level privacy benchmark to pair the same real image with matched private and safe social-media captions. It contains 10,398 accessible, real-world images from VISPR, PrivacyAlert, and DIPA2. Each caption pair describes the same broad context in a plausible first-person style. Private captions are labeled with exact textual evidence and disclosure provenance. Dataset-specific labels are mapped to a shared 14-category privacy taxonomy, yielding 20,796 captions for controlled cross-modal evaluation.

Our methodology retains three established tasks for binary detection, taxonomy-guided classification, and multi-label recognition, and adds a fourth task that predicts image and caption privacy separately. An image-only replication with four open-weight vision-language models reveals substantial class bias, prompt sensitivity, and uneven coverage of privacy categories. By varying caption privacy while holding the image and posting context fixed, PRIVCAP enables systematic measurement of caption effects, category coverage, and disclosure-source attribution.

## 1 Introduction

Privacy on social media is a property of the complete post, not of either modality in isolation. An image may reveal a face, document, vehicle, or physical setting, while its caption can state a name, relationship, location, institution, routine, or life event that is not visually recoverable. The same photograph can therefore carry different privacy risks under different captions. Evaluating images alone misses this interaction, while evaluating an arbitrary image-caption pair cannot determine whether a change in prediction was caused by the caption or by a different visual context.

Visual-privacy benchmarks remain predominantly image-level. VISPR Orekondy et al. [2017] introduced 68 fine-grained privacy attributes, and Tsaprazlis et al. [2025] evaluated VLMs across VISPR, PrivacyAlert, and DIPA2 using binary detection, taxonomy-guided classification, attribute recognition, and caption augmentation. Other multimodal studies use tags or focus on a single risk such as geolocation Tonge and Caragea [2019], Wang et al. [2025]. These settings do not hold the image and broad posting context fixed while varying whether the caption contains a disclosure, nor do they provide separate targets for the image and caption channels.

Naturally authored private captions would provide realism, but collecting and redistributing them raises consent and data-protection concerns. Following evidence that generated text can support

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**Figure placeholder**

Insert the PrivCap overview diagram: source image and annotations → dataset-specific annotation verification → single-pass caption and label generation → private and safe caption pair with label evidence and provenance → evaluation-ready example. Show one abstract sample without identifiable personal content.

Figure 1: **Overview of PrivCap.** Each source image is paired with contrastive private and safe captions. Labels attached to the private caption include exact textual evidence and modality provenance.

privacy research Yukhymenko et al. [2024], we retain real, non-synthetic benchmark images and generate matched captions. Both captions describe the same broad moment in a natural first-person social-media style. The private caption states at least one target disclosure, while the safe caption adds none. This construction changes caption privacy without changing the underlying image or the general posting scenario.

We introduce **PrivCap**, comprising 10,398 accessible images from VISPR, PrivacyAlert, and DIPA2 and 20,796 matched captions. Caption generation has access to each dataset's complete fine-grained privacy-label vocabulary, allowing the text to express privacy categories beyond those visible in its paired image. Each private-caption label is linked to an exact evidence phrase and marked as image-grounded or caption-induced. The dataset-specific labels are then mapped to a shared 14-category taxonomy that covers both visual and textual disclosures.

Our evaluation separates visual-only capability from post-level privacy understanding. We reproduce three established image-only tasks on four newer open-weight VLMs, then apply the same tasks to private- and safe-caption conditions. A new fourth task asks for separate Private/Safe judgments for the image and caption, directly testing source attribution. The image-only results already expose an important methodological issue: aggregate scores can conceal severe class bias, and taxonomy guidance can rebalance one model while pushing another toward false positives. Fine-grained recognition also remains uneven across common and rare privacy categories. These findings motivate controlled caption comparisons and channel-specific evaluation rather than reliance on a single post-level score.

**Contributions.**

1. **A controlled post-level privacy benchmark.** PrivCap pairs 10,398 real-world images from three established benchmarks with private and safe first-person social-media captions, producing 20,796 matched caption conditions.
2. **Cross-modal privacy annotations.** Fine-grained caption labels include exact textual evidence and disclosure provenance, and map with source-image labels to a shared 14-category taxonomy spanning visual and textual privacy.
3. **Evaluation of caption effects and source attribution.** The methodology combines three established tasks with a new task that scores image and caption privacy separately, supporting analysis of class bias, category coverage, caption effects, and cross-modal source confusion.

## 2 Related Work

### 2.1 Visual Privacy Benchmarks

VISPR Orekondy et al. [2017] provides over 22,000 images annotated with 68 fine-grained privacy attributes spanning identity, health, location, relationships, and digital identifiers. PrivacyAlert Zhao et al. [2022] supplies binary private/safe labels with a strong safe-class majority, while DIPA2 Li et al. [2023] provides object-level privacy annotations but no safe examples. Their complementary annotation granularity and class balance motivate our three-dataset evaluation.

Tsaprazlis et al. [2025] mapped VISPR attributes to a 14-category hierarchy derived from GDPR European Parliament [2016] and CCPA California State Legislature [2018], then evaluated five VLMs across VISPR, PrivacyAlert, and DIPA2 using binary detection, taxonomy-guided detection, attribute recognition, and caption augmentation. We adopt their taxonomy and replicate the three image-only tasks on four newer models, while replacing generic caption augmentation with matched, taxonomy-grounded private and safe captions.

### 2.2 Social-Media and Multimodal Privacy

Social-media privacy has been studied through PII extraction Liu et al. [2021], multi-label disclosure detection in Twitter posts Liang et al. [2024], and personal-attribute inference from text Staab et al. [2024]. These methods analyze the textual channel alone and cannot determine whether a disclosure originates from an image, caption, or both.

Multimodal work considers related but different settings. Image tags support privacy prediction in Tonge and Caragea [2019], while MultiTrust Zhang et al. [2024b] and Multi-PA Zhang et al. [2024a] use textual queries or image-question pairs. KoreaGEO Wang et al. [2025] studies image-only and caption-conditioned geolocation but is restricted to location privacy. Unsafe2Safe Dinh and Jin [2026] uses paired private and public captions for image anonymization rather than privacy detection.

PRISM Li et al. [2025] generates synthetic profiles, posts, and images to evaluate inference of 12 private attributes from multimodal user histories. SopriBench Peng et al. [2026] similarly measures cumulative leakage across synthetic social-media profiles. Both target user-level profiling from multiple posts. PRIVCAP instead evaluates individual posts built from real-world, non-synthetic images and matched private/safe captions, holding the image and broad posting context fixed. It further covers 68 fine-grained labels mapped to 14 categories and evaluates separate Private/Safe decisions for the image and caption channels.

Synthetic text avoids reproducing naturally authored disclosures. SynthPAI Yukhymenko et al. [2024] showed that generated Reddit-style text can support personal-attribute inference research. PRIVCAP adopts this ethical advantage for captions while retaining real images; its generated captions are image-consistent, written as plausible first-person social-media posts, and paired contrastively for controlled evaluation.

## 3 The PRIVCAP Dataset and Evaluation Framework

### 3.1 Privacy Taxonomy

PRIVCAP uses the 68-label VISPR taxonomy Orekondy et al. [2017], which covers privacy disclosures in both images and captions. This shared label space supports cross-modal comparison and modality provenance. Evaluation maps these labels to the 14-category GDPR/CCPA-derived taxonomy of Tsaprazlis et al. [2025]; Appendix A provides the complete mapping.

### 3.2 The PRIVCAP Generation Pipeline

PRIVCAP pairs real-world, non-synthetic images with generated captions. Complete pairs are available for 7,995 VISPR, 1,550 PrivacyAlert, and 853 DIPA2 images. DIPA2 evaluation uses the 265 images with unanimous agreement across all four annotators.

**Source annotation verification.** Dataset-specific pipelines recheck image annotations using structured Gemini outputs at temperature zero. VISPR attributes are verified against the full image; PrivacyAlert private/public decisions and native categories are rechecked; and DIPA2 object categories are tested for visible support. PrivacyAlert and DIPA2 use a 0.60 confidence threshold. DIPA2 uses padded annotation crops when available and otherwise the full image; no new boxes are generated. Each run stores verified labels, rationales, and confidence values.

**Single-pass caption and label generation.** Given the image, source metadata, and privacy-label guide, one call produces a private caption, a matched safe caption, and private-caption labels. The guide contains the dataset's complete fine-grained privacy-label vocabulary, not only the labels associated with the current image. A private caption may therefore introduce a disclosure absent from the paired image while remaining within the dataset's label space. Both captions are plausible first-person posts describing the same broad moment. The private caption contains at least one disclosure; the safe caption adds none. Every private-caption label includes an exact evidence substring and `image_grounded` or `caption_induced` provenance. Safe captions receive no privacy labels.

**Automatic output validation.** Records are retained only when both captions, valid labels, and label evidence are present. The parser normalizes labels and logs failures for regeneration.

| Privacy category | VISPR Image n=38,874 | VISPR Private caption n=19,200 | PrivacyAlert Image n=2,283 | PrivacyAlert Private caption n=838 | DIPA2 Image n=1,094 | DIPA2 Private caption n=2,135 |
|---|---|---|---|---|---|---|
| Financial Information | **0.3%** (121) | **1.4%** (269) | **0.0%** (0) | **0.1%** (1) | - | - |
| Nudity | **1.2%** (460) | **0.9%** (166) | **28.6%** (654) | **6.1%** (51) | - | - |
| Vehicle Information | **0.7%** (271) | **2.5%** (480) | **2.1%** (49) | **2.3%** (19) | **8.8%** (96) | **4.5%** (97) |
| HIPAA Data | **0.6%** (225) | **2.6%** (507) | **3.4%** (78) | **2.0%** (17) | - | - |
| Location Identifiers | **3.3%** (1,298) | **12.3%** (2,366) | **0.0%** (1) | **0.0%** (0) | **9.3%** (102) | **13.0%** (277) |
| Personal Context | **6.2%** (2,428) | **31.3%** (6,002) | **15.2%** (348) | **32.6%** (273) | **34.3%** (375) | **37.7%** (804) |
| Digital Identifiers | **0.5%** (209) | **1.2%** (239) | **5.3%** (121) | **3.2%** (27) | **10.8%** (118) | **15.9%** (339) |
| Personal Metadata (Demographics) | **33.6%** (13,081) | **25.8%** (4,953) | **11.0%** (250) | **19.2%** (161) | - | - |
| Background Individuals | **1.5%** (565) | **1.5%** (282) | **7.5%** (171) | **3.6%** (30) | **0.9%** (10) | **0.8%** (17) |
| Legal Identifiers | **5.6%** (2,181) | **13.9%** (2,671) | **3.1%** (70) | **11.0%** (92) | **4.8%** (52) | **4.7%** (100) |
| Biometric Data | **46.3%** (18,015) | **4.2%** (807) | **8.8%** (201) | **5.7%** (48) | **29.6%** (324) | **22.2%** (475) |
| Violent/Unlawful Actions | **0.1%** (20) | **2.4%** (458) | **11.1%** (253) | **8.5%** (71) | **1.6%** (17) | **1.2%** (26) |
| Children | - | - | **3.8%** (87) | **5.7%** (48) | - | - |

Image Private caption Cell: percentage (raw count).

Figure 3: **Privacy categories by modality.** Each mapped source-image label (blue) and explicit private-caption label (orange) contributes once to its category in the taxonomy of Tsaprazlis et al. [2025]. Caption labels are counted whether or not their category occurs in the paired image. Percentages are normalized over all mapped labels within each dataset and modality, so each column totals 100% before rounding; parentheses show label counts. Safe is excluded.

### 3.3 Dataset Statistics

PRIVCAP contains 10,398 images and 20,796 captions, one private and one safe caption per image. VISPR and PrivacyAlert contain private and safe image targets; DIPA2 contains only private images. Every dataset contains both caption conditions, supporting separate modality targets wherever source labels permit. Table 1 reports the complete corpus counts.

Table 1: **PRIVCAP corpus statistics.**

| Source | Total images | Private images | Safe/Public images | Private captions | Safe captions | Total captions |
|---|---|---|---|---|---|---|
| VISPR | 7,995 | 4,989 | 3,006 | 7,995 | 7,995 | 15,990 |
| PrivacyAlert | 1,550 | 366 | 1,184 | 1,550 | 1,550 | 3,100 |
| DIPA2 | 853 | 853 | 0 | 853 | 853 | 1,706 |
| **Total** | **10,398** | **6,208** | **4,190** | **10,398** | **10,398** | **20,796** |

**Figure placeholder**

Insert the evaluation diagram: image-only → T1–T3; image with matched private/safe caption → T1–T4; T4 → separate image-private and caption-private outputs.

Figure 2: **Evaluation methodology.** Image-only input uses T1–T3; caption-conditioned input uses T1–T4, with separate T4 judgments for each modality.

**Privacy categories by modality.** Figure 3 compares the category composition of source-image annotations and explicit private-caption labels. Across datasets, image labels concentrate on visually observable attributes, whereas captions place greater weight on contextual and identifying information that can be stated explicitly. Personal Context is the largest caption category in all three datasets. VISPR images are dominated by Biometric Data, while its captions emphasize Personal Context, Personal Metadata, Legal Identifiers, and Location Identifiers. PrivacyAlert shows the clearest shift from image-side Nudity to caption-side Personal Context, Personal Metadata, and Legal Identifiers. DIPA2 captions retain a strong Personal Context component while assigning greater relative weight to Digital and Location Identifiers. Section 5.1 provides the detailed analysis.

### 3.4 Evaluation Methodology

We retain three tasks from Tsaprazlis et al. [2025] and add one joint task.

**Tasks.**

**T1: Binary Privacy Detection.** Assigns one Private/Safe label using a concise privacy-detection prompt.

**T2: Taxonomy-Guided State Classification.** Assigns the same joint Private/Safe label with the 14-category taxonomy included in the prompt.

**T3: Multi-Label Attribute Recognition.** Identifies the complete set of privacy categories present in the input.

**T4: Joint Taxonomy-Guided Source Attribution.** Predicts Private/Safe independently for the image and caption within one taxonomy-guided prompt, testing whether the model attributes disclosure to the correct modality.

**Input conditions.** Image-only input reproduces T1–T3. Multimodal input pairs the same image with either its private or safe caption and applies T1–T4, isolating privacy-bearing text while preserving the broad context.

**Multimodal ground truth.** For T1–T2, private-caption pairs are Private; safe-caption targets follow the image label. T3 uses the union of mapped image and caption-induced categories for private captions and image categories alone for safe captions. T4 image targets use mapped image annotations; caption targets are positive for caption-induced disclosure and negative for safe captions. Image-grounded caption labels are not counted twice.

## 4 Experimental Setup

### 4.1 Models

We evaluate four open-weight VLMs spanning 3–11B parameters, including LLaMA 3.2-11B as a replication anchor.

**Model specifications.** Checkpoints and image preprocessing used in evaluation.

| Model | Params | HuggingFace ID | Precision | Max px |
|---|---|---|---|---|
| Ministral-3B | 3B | `mistralai/Ministral-3-3B-Instruct-2512-BF16` | bfloat16 | 1,024 |
| Gemma-3-4B | 4B | `google/gemma-3-4b-it` | bfloat16 | 1,024 |
| Qwen3-VL-8B | 8B | `Qwen/Qwen3-VL-8B-Instruct` | bfloat16 | 1,024 |
| LLaMA-3.2-11B | 11B | `meta-llama/Llama-3.2-11B-Vision-Instruct` | bfloat16 | 1,024 |

Images are resized to at most 1,024 pixels on the longest edge. Each model runs three fixed seeds at $T \in \{0.1, 1.0\}$; majority vote determines binary outputs and retains T3 labels appearing in at least two runs. T4 votes independently by channel, and the best Macro-F1 temperature is reported per model and task Tsaprazlis et al. [2025]. Runs use NVIDIA A100 GPUs in Google Colab.

### 4.2 Datasets

**VISPR** Orekondy et al. [2017]: we use 7,995 accessible images with completed caption pairs. Any non-`safe` attribute makes an image private.

**PrivacyAlert** Zhao et al. [2022]: image-only runs use 1,554 accessible test images (370 private; 1,184 public); caption-conditioned runs use 1,550 completed pairs (366 private; 1,184 public).

**DIPA2** Li et al. [2023]: generation covers 853 private images, while evaluation uses the 265-image unanimous-agreement subset. Image-only results currently report T3; caption-conditioned binary tasks use private-class recall because no safe images are available.

### 4.3 Metrics

Binary tasks report Macro F1 on VISPR and PrivacyAlert, with Accuracy added for image-only T1; DIPA2 uses private-class recall. T3 reports per-category and macro Precision, Recall, and F1. T4 reports positive-class metrics per channel, except safe-caption behavior uses caption false-positive rate because it has no positive caption targets. Caption effect is $\Delta\text{F1} = \text{F1}_{\text{private}} - \text{F1}_{\text{safe}}$, with the analogous recall difference for binary DIPA2 tasks.

### 4.4 Baselines

Image-only T1–T3 replicate the evaluation methodology of Tsaprazlis et al. [2025], but the resulting scores are not directly comparable. Our accessible dataset subsets differ from theirs because some source links were unavailable, including 246 PrivacyAlert images, which changes the evaluation split. We also evaluate a newer set of VLMs. Caption-conditioned T1–T4 constitute our proposed evaluation.

## 5 Results and Analysis

### 5.1 Privacy Categories by Modality

Figure 3 reveals modality-specific category profiles whose magnitude varies by dataset. VISPR image labels concentrate on Biometric Data (46.3%) and Personal Metadata (33.6%), whereas caption labels concentrate on Personal Context (31.3%), Personal Metadata (25.8%), Legal Identifiers (13.9%), and Location Identifiers (12.3%). PrivacyAlert image labels are led by Nudity (28.6%), while caption labels shift toward Personal Context (32.6%), Personal Metadata (19.2%), and Legal Identifiers (11.0%). In DIPA2, Personal Context is largest in both modalities. Relative to images, captions allocate more share to Digital and Location Identifiers, while images allocate more to Biometric Data and Vehicle Information. Aggregate coverage does not imply pairwise overlap: 572 of 853 DIPA2 pairs (67.1%) contain at least one caption category absent from the paired image annotations.

This division is consistent with how social-media posts convey private information. Images directly encode faces, bodies, vehicles, documents, and surrounding scenes, whereas captions state relationships, routines, demographics, locations, institutions, and digital activity. The observed profiles therefore reflect modality-specific disclosure channels and dataset composition, not an inherent privacy advantage for either modality.

### 5.2 Visual-Only Privacy Benchmark Replication

Table 2 reports best-temperature Macro F1 for T1–T3 using matched prompts, decoding, parsing, and metrics. VISPR scores are computed on the 7,995 accessible, caption-complete image IDs. Only our accessible splits are included because PrivacyAlert link loss prevents direct score comparison.

Table 2: **Visual-only benchmark replication.** Best-temperature Macro F1 (%). PA denotes PrivacyAlert; bold marks each column maximum.

| | *T1: Binary Detection* | | *T2: Taxonomy-Guided* | | *T3: Attribute Recognition* | | |
|---|---|---|---|---|---|---|---|
| **Model** | **VISPR** | **PA** | **VISPR** | **PA** | **VISPR** | **PA** | **DIPA2** |
| Ministral-3B | 27.74 | 44.61 | **64.41** | **73.18** | 34.76 | **15.13** | 26.38 |
| Gemma-3-4B | **63.65** | **67.96** | 63.36 | 62.22 | 39.07 | 9.33 | 34.54 |
| Qwen3-VL-8B | 32.82 | 52.84 | 41.32 | 65.31 | **47.90** | 14.09 | **40.58** |
| LLaMA-3.2-11B | 55.81 | 44.11 | 44.41 | 69.21 | 31.06 | 9.70 | 24.30 |

The aggregate scores conceal pronounced class-specific behavior. Under the concise T1 prompt, Ministral and Qwen default largely to Safe: their Private recall is 0.40% and 5.21% on VISPR and 1.35% and 10.00% on PrivacyAlert, while Safe recall remains at least 99.60%. Their comparatively high PrivacyAlert Accuracy is therefore driven by the majority class rather than balanced privacy detection. Gemma is the only model with comparatively balanced T1 recall on both datasets, while LLaMA is more balanced on VISPR than on PrivacyAlert.

Taxonomy guidance changes model behavior but is not uniformly beneficial. In T2, Ministral's Private recall rises to 49.17% on VISPR and 58.65% on PrivacyAlert without discarding strong Safe recall, producing the highest Macro F1 on both datasets. Qwen remains Safe-biased, whereas

Gemma shifts toward Private predictions on PrivacyAlert, reaching 91.35% Private recall but only 38.90% precision and 55.15% Safe recall. T3 exposes a separate coverage problem. On VISPR, even the strongest model attains only 19.34% F1 for Biometric Data despite its large support; on PrivacyAlert, every model records zero F1 for several sparse categories despite substantially better Nudity recognition; and on DIPA2, Qwen's leading Macro F1 coexists with zero F1 for Legal Identifiers and Violent/Unlawful Actions. Prompt context can therefore rebalance or overcorrect binary decisions, while aggregate binary performance does not establish broad privacy-category recognition.

### 5.3 Cross-Modal Privacy Evaluation

Table 3 compares the same image with private and safe captions. T1–T2 use Macro F1 for VISPR/PrivacyAlert and private-class recall for DIPA2; T3 uses Macro F1. Table 4 separates the two attribution channels.

Table 3: **Cross-modal T1–T3 evaluation.** Private- and safe-caption conditions are reported separately. T1–T2 use Macro F1 for VISPR/PrivacyAlert and private recall for DIPA2; T3 uses Macro F1. All values are percentages.

| | *T1: Binary Detection* | | | *T2: Taxonomy-Guided* | | | *T3: Attribute Recognition* | | |
|---|---|---|---|---|---|---|---|---|---|
| **Model** | **VISPR** | **PA** | **DIPA2** | **VISPR** | **PA** | **DIPA2** | **VISPR** | **PA** | **DIPA2** |
| **Image + private caption** | | | | | | | | | |
| Ministral-3B | - | - | - | - | - | - | - | - | - |
| Gemma-3-4B | - | - | - | - | - | - | - | - | - |
| Qwen3-VL-8B | - | - | - | - | - | - | - | - | - |
| LLaMA-3.2-11B | - | - | - | - | - | - | - | - | - |
| **Image + safe caption** | | | | | | | | | |
| Ministral-3B | - | - | - | - | - | - | - | - | - |
| Gemma-3-4B | - | - | - | - | - | - | - | - | - |
| Qwen3-VL-8B | - | - | - | - | - | - | - | - | - |
| LLaMA-3.2-11B | - | - | - | - | - | - | - | - | - |

Table 4: **T4 source attribution.** Image columns report positive-class F1 by caption condition; caption columns report private-caption F1 and safe-caption false-positive rate. All values are percentages.

| **Dataset** | **Model** | **Img-P F1** | **Img-S F1** | **Cap-P F1** | **Cap-S FPR** |
|---|---|---|---|---|---|
| VISPR | Ministral-3B | - | - | - | - |
| | Gemma-3-4B | - | - | - | - |
| | Qwen3-VL-8B | - | - | - | - |
| | LLaMA-3.2-11B | - | - | - | - |
| PrivacyAlert | Ministral-3B | - | - | - | - |
| | Gemma-3-4B | - | - | - | - |
| | Qwen3-VL-8B | - | - | - | - |
| | LLaMA-3.2-11B | - | - | - | - |
| DIPA2 | Ministral-3B | - | - | - | - |
| | Gemma-3-4B | - | - | - | - |
| | Qwen3-VL-8B | - | - | - | - |
| | LLaMA-3.2-11B | - | - | - | - |

T1–T3 measure within-image caption effects; T4 measures correct source assignment and safe-caption false positives.

### 5.4 Model-Level Analysis

The final analysis will compare image-only detection, caption sensitivity, category recognition, and source attribution.

Table 5: **Model-level analysis summary.** Values will be entered after the final evaluation runs.

| **Model** | **Image-only T1** | **Caption effect T1–T2** | **T3 recognition** | **T4 attribution** |
|---|---|---|---|---|
| Ministral-3B | - | - | - | - |
| Gemma-3-4B | - | - | - | - |
| Qwen3-VL-8B | - | - | - | - |
| LLaMA-3.2-11B | - | - | - | - |

### 5.5 Error Analysis

We will audit stored targets, majority votes, run-level predictions, and raw outputs using the categories below.

Table 6: **Error-analysis template.** Counts and representative example identifiers will be filled from the final predictions.

| **Error type** | **Definition** | **Count** | **Rate** | **Example IDs** |
|---|---|---|---|---|
| Missed image disclosure | Private image evidence predicted Safe or absent | - | - | - |
| Missed caption disclosure | Private-caption evidence predicted Safe or absent | - | - | - |
| Caption false positive | Safe caption predicted Private | - | - | - |
| Source confusion | Correct joint decision but incorrect T4 source channel | - | - | - |
| Taxonomy mismatch | Privacy detected under the wrong privacy category | - | - | - |
| Parsing failure | Output cannot be mapped to the requested schema | - | - | - |

## 6 Conclusion

We introduced PrivCap, a controlled post-level benchmark pairing 10,398 real-world images with 20,796 private and safe social-media-style captions. Its evidence-linked privacy labels include modality provenance and map from 68 fine-grained labels to 14 evaluation categories. The methodology separates image-only replication from caption-conditioned evaluation and adds joint taxonomy-guided source attribution.

**Limitations.**

PrivCap uses English captions from one generator. Despite image and style constraints, they may reflect its linguistic and cultural biases rather than the diversity of naturally authored posts. Other languages, platforms, caption styles, and generators remain untested.

PrivacyAlert is class-imbalanced, and DIPA2 lacks safe examples. Model-assisted verification is not exhaustive human relabeling, so annotation errors may remain. Fixed privacy categories also cannot fully represent contextual integrity Nissenbaum [2004], which depends on audience, purpose, platform, and expected information flow.

**Future work.** Future work will study adversarial and ambiguous captions, naturally authored and multilingual posts, and video. We will release PrivCap, verified annotations, caption pairs, and generation and evaluation code.

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# A Source and Verified Label Counts

Table 7 reports category-level source and retained image-grounded counts. *Source* denotes images with at least one raw source label or keyword mapped to a category. *Verified* denotes images that retain at least one mapped image-grounded label after the dataset-specific verification or grounding step. Each image contributes at most once to a category in each column. These counts must not be confused with the complete private-caption label counts in Figure 3.

The available VISPR audit rollup covers 1,000 images. The DIPA2 columns use 853 retained generation records. For PrivacyAlert, we reconstruct the rollup from all 1,550 generation JSONs: Source uses the aggregated source keywords, while Verified uses labels retained with `image_keyword` or `image_grounded` provenance after image-conditioned generation. Because the generator selects a concise evidence-bearing set, the PrivacyAlert Verified column measures retained grounded category presence rather than exhaustive relabeling. A verified binary Safe decision is not recoverable from these records and remains unreported.

Table 7: **Source and verified image-label counts.** Per-image category presence before and after dataset-specific verification or grounding. VISPR uses the available 1,000-image audit, PrivacyAlert uses 1,550 generation records, and DIPA2 uses 853 retained generation records.

| **Evaluation category** | **VISPR** | | **PrivacyAlert** | | **DIPA2** | |
|---|---|---|---|---|---|---|
| | **Source** | **Verified** | **Source** | **Verified** | **Source** | **Verified** |
| Biometric Data | 501 | 488 | 186 | 47 | 319 | 319 |
| Children Images | 0 | 0 | 74 | 44 | 0 | 0 |
| Financial Information | 25 | 22 | 0 | 0 | 0 | 0 |
| HIPAA Data | 22 | 19 | 78 | 15 | 0 | 0 |
| Legal Identifiers | 178 | 131 | 70 | 8 | 52 | 52 |
| Digital Identifiers | 24 | 21 | 91 | 23 | 115 | 115 |
| Personal Metadata (Demographics) | 507 | 492 | 172 | 98 | 0 | 0 |
| GPS Data | 0 | 0 | 0 | 0 | 0 | 0 |
| Vehicle Information | 30 | 24 | 49 | 16 | 96 | 96 |
| Nudity | 53 | 49 | 392 | 50 | 0 | 0 |
| Violent/Unlawful Actions | 3 | 3 | 196 | 68 | 17 | 17 |
| Personal Context | 221 | 213 | 268 | 153 | 336 | 336 |
| Location Identifiers | 157 | 131 | 1 | 0 | 98 | 98 |
| Background Individuals | 80 | 78 | 171 | 30 | 10 | 10 |
| Safe | 348 | 329 | 1,184 | - | 0 | 0 |

## A.1 Dataset-Specific Mapping to the Evaluation Taxonomy

Table 8 gives the native privacy labels or keywords mapped to each evaluation category for VISPR, PrivacyAlert, and DIPA2. This is the mapping used to construct the category targets and the modality profiles in Figure 3. Labels are shown in normalized display form; matching is case-insensitive and treats spaces, hyphens, and underscores equivalently.

Table 8: **Dataset-specific privacy-label mapping.** Native dataset labels mapped to the shared evaluation taxonomy.

| Eval. Category | VISPR | PrivacyAlert | DIPA2 |
|---|---|---|---|
| Biometric Data | face complete, height approx, eye color, tattoo, hair color, weight approx, face partial, color, fingerprint | eye, tattoo, ungroomed, messy hair, overweight, piercing, unflattering appearance, unsatisfying body, unfashionable, funny looking, strange hair, wig, scary looking, silly, forced smile, unamused face | Person, Finger |
| Children Images | - | children, boy, kids, children in danger | - |
| Financial Information | receipt, credit card | bank account, credit card | - |
| HIPAA Data | injury, physical disability, medicine | abscess, peeling skin, bad teeth, acne, bloody, injury, wound, surgery, peel, pharmacy, emergency, tongue, gummy, throat, lip, infection, pain | - |
| Legal Identifiers | last name, passport, birth city, complete home address, handwriting, mail, partial current address, partial home address, identity card, first name, birth date, complete current address, driver's license, signature, ticket, full name, student ID | home address, passport, sign | Identity, Printed Material, Printed Materials |
| Digital Identifiers | email content, online conversation, username, email, phone | email, username, password, ticket, laptop, browser, computer, internet, railway, flight | Machine, Screen, Electronic Devices |
| Personal Metadata (Demographics) | race, religion, shared views, general opinion, nationality, approximate age, sexual orientation, ethnic clothing, marital status, political opinion, culture, gender, occupation | culture, religion, spiritual, bible, catholic, christian, christianity, church, faith, hinduism, indian, islam, judaism, religious, traditional | - |
| GPS Data | - | - | - |
| Vehicle Information | vehicle ownership, complete license plate, partial license plate | license plate, automobile | Vehicle Plate |
| Nudity | full nudity, semi-nudity | bare, body, breasts, butt, erotic, naked, nudity, sexual, shirtless, bra, bikini | - |

| Eval. Category | VISPR | PrivacyAlert | DIPA2 |
|---|---|---|---|
| Violent/Unlawful Actions | legal involvement | damage, guns, war, military, shooting, weapons, corps, battlefield, smoking, alcohol, cigarette, illegal, drugs, arrest, mugshot, marijuana, drug, kills, thief, bomber, smuggler, prisons, robbery | Cigarettes |
| Personal Context | personal occasion, sports, work occasion, education history, hobbies, personal relationship, competitors, professional relationship, social relationship | family, husband, parents, partner, wife, drunk, hang out, party, music, event, concert | Clothing, Book, Table, Toy, Home Interior, Cosmetics, Musical Instrument, Accessory, Pet, Food |
| Location Identifiers | date and time, landmark | landmark | Place Identifier, Scenery |
| Background Individuals | spectators | spectator, people | Photo |
| Safe | safe | public | - |

# B Additional Dataset and Result Tables

## B.1 Prompt and Parsing Specification

In the prompts below, `[caption]` is replaced with the paired caption and `[taxonomy]` expands to the complete evaluation taxonomy with category examples. The wording and output schemas follow the multimodal evaluation notebooks.

Table 9: **Task prompts and response processing.** Exact task instructions with repeated taxonomy text represented by `[taxonomy]`.

| Task | Prompt | Output and parsing |
|---|---|---|
| T1 binary privacy detection | *You are given a social media post consisting of an image and an accompanying caption. Caption: "[caption]". Considering both the image and the caption, identify if this post violates any privacy policies. If yes, say "Yes"; otherwise say "No".* | `Yes` or `No`. Case-insensitive binary matching is used; unrecognized output maps to Safe. |
| T2 taxonomy-guided state classification | *You are given a social media post consisting of an image and an accompanying caption. Caption: "[caption]". Identify whether this post violates any privacy policy in [taxonomy]. If yes, classify it as Private; otherwise classify it as Safe. Do not justify the answer. Output only the requested JSON.* | `{"privacy_state": ["Private"]}` or `{"privacy_state": ["Safe"]}`. JSON is parsed first, followed by Private/Safe keyword matching; unrecognized output maps to Safe. |

| Task | Prompt | Output and parsing |
|---|---|---|
| T3 multi-label attribute recognition | *You are given a social media post consisting of an image and an accompanying caption. Caption: "[caption]". Identify all privacy categories from [taxonomy] that apply to the image or caption. Use exact category names, do not justify the answer, and output only the requested JSON.* | `{"categories": [...]}` or `{"categories": ["Safe"]}`. JSON labels are resolved to exact taxonomy names, followed by exact-name text matching; Safe or unrecognized output maps to an empty set. |
| T4 source attribution | *You are given a social media post consisting of an image and an accompanying caption. Caption: "[caption]". Using [taxonomy], determine whether the image alone contains private information and whether the caption alone contains private information. Answer each channel with Yes or No, do not justify the answer, and output only the requested JSON.* | `{"image_private": "Yes/No", "caption_private": "Yes/No"}`. JSON is parsed first, followed by a source-specific line scan; a missing channel maps to No. |

**Dataset Governance and Redistribution**

Table 10: **Dataset governance and redistribution.** Provenance, usage terms, and privacy safeguards for each source asset.

| Source | License or terms | Version or access date | Redistribution | Safeguards |
|---|---|---|---|---|
| VISPR | - | - | - | - |
| PrivacyAlert | - | - | - | - |
| DIPA2 | - | - | - | - |
| Generated captions | - | - | - | - |
| Model outputs | - | - | - | - |

# C Implementation Details

## C.1 Code and Data Availability

Code, derived annotations, caption pairs, and reproducibility instructions will be provided in an anonymized repository:

`https://github.com/anonymous-author/privcap`

The repository will include scripts for obtaining and validating the permitted source images, jointly generating private and safe captions with their privacy labels and evidence phrases, running T1–T4, and reconstructing every reported table from per-example predictions. A versioned environment file and a README will specify the exact installation, data-preparation, inference, and evaluation commands. Source images will be distributed only where permitted by the original dataset terms; otherwise, the repository will provide identifiers and acquisition instructions.

**Generation and Evaluation Configuration**

**Compute environment.** All model evaluation runs were executed in Google Colab using NVIDIA A100 GPUs.

**Source annotation verification.** VISPR attributes and PrivacyAlert binary/category annotations are verified with `gemini-3.1-flash-lite` at $T{=}0$. VISPR removes labels judged absent and conservatively retains labels without a returned verdict. PrivacyAlert changes a source binary label or existing category only at confidence $\geq$ 0.60. DIPA2 object categories are verified with `gemini-2.0-flash` at $T{=}0$ and retained when visible with confidence $\geq$ 0.60; per-item errors are retained conservatively. Existing DIPA2 manual boxes may define padded crops for verification, but no new bounding boxes are produced.

**Caption generation.** Private and safe caption pairs, private-caption labels, evidence phrases, and provenance are produced in one call per image at $T{=}0.7$. VISPR and DIPA2 records identify `gemini-3.1-flash-lite-preview`; PrivacyAlert records identify `gemini-3.1-flash-lite`. Processing is resumable, existing output files are skipped, failures are logged separately, and checkpoints are written every 500 images.

**Model evaluation.** The evaluation notebooks use HuggingFace model-specific generation interfaces and resize images to a maximum longest edge of 1,024 pixels. The saved configurations evaluate $T \in \{0.1, 1.0\}$ with seeds $\{0, 42, 84\}$ and three generations per example. Binary outputs are decided by majority vote; a T3 category is retained when predicted in at least two runs; and T4 image and caption judgments are voted independently. The best-temperature result is selected by the task's reported F1 measure. Model identifiers and precision are reported in Section 4.

**Validation.** Each run stores per-example ground truth, parsed predictions, all three run-level predictions, and raw model outputs. Final metrics are recomputed from these records so that prompt parsing, majority voting, and aggregate scores can be independently checked.